CLASS - XII PASS [LEADRER]

CT – 1 [NEET]

Date : 15.07.2023

Instructions

1. The Test consists of **180** questions with **3** Sections: **Section I-PHYSICS** , **Section II-CHEMISTRY** , **Section III-BIOLOGY (PART – I BOTANY** & **PART – II ZOOLOGY)** .
2. Duration of Test is **3hrs 20 minutes.** and Questions Paper Contains **200** Questions , **50** questions in each section **(attempt only 45) Max. Marks 720** .
3. Each correct answer carries **4 marks** , while **1 mark will be deducted for every wrong answer.**
4. Student cannot use log tables and calculators or any other electronic material in the examination hall.
5. Student must abide by the instructions issued during the examination, by the invigilators or the centre in charge.
6. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
7. A candidate has write his/her answer in the OMR sheet by darkening the appropriate bubble with the help of **Blue/Black ball Point Pen only** as the correct answer(s) of the question attempted.
8. **Use of Pencil and eraser is strictly prohibited.**
9. Must collect answer key after the examination is held.
10. Do not leave examination hall related to any type of error in question paper .It shall be managed by examination in-charge after the discussion with the subject teacher.

SYLLABUS

PHYSICS	: Kinematics (Motion in ID), Vector, Thermal Expansion, Heat Transfer
CHEMISTRY	: Stoichiometry, Structure of Atom, Nomenclature, Isomerism
BOTANY	: Living World, Biological Classification (Monera, Protista and Fungi)
ZOOLOGY	: Animal Kingdom (Complete)

Student Name

Enrollment No.

PART-I PHYSICS

Section A

Sanjeev sir Ques

1. A body moving with a uniform acceleration has velocities of u and v when passing through points A and B in its path. The velocity of the body midway between A and B is

(a) $(u + v) / 2$ (b*) $\frac{u^2 + v^2}{2}$

(c) uv (d) None of these

7 Ex - 2

2. In a car race car A takes t_0 time less to finish than car B and passes the finishing point with velocity v_0 more than car B. The cars start from rest and travel with constant acceleration a_1 and a_2. Then the ratio of v_0/t_0 is equal to

(a) $\frac{a_2^2}{a_1}$ (b) $(a_1 + a_2) / 2$

(c*) a_1a_2 (d) $\frac{a_2^2}{a_1}$

9

3. A stone is thrown vertically upward with an initial velocity v_0. The distance travelled in time $4v_0 / 3g$ is

(a) $\frac{2v_0^2}{g}$ (b) $\frac{v_0^2}{2g}$

(c) $\frac{4v_0^2}{3g}$ (d*) $\frac{5v_0^2}{9g}$

5 Ex-3

4. A ball is dropped from the roof of a tower of height h. The total distance covered by it in the last second of its motion is equal to the distance covered by it in first 3 seconds. The value of h in meters is ($g = 10 m/s^2$)

(a*) 125 (b) 200

(c) 100 (d) 80

7

5. If x denotes displacement in time t and $x = a \cos t$, then acceleration is

(a) $a \cos t$ (b*) $- a \cos t$

(c) $a \sin t$ (d) $- a \sin t$

9

6. A bird moves from point (1, -2, 3) to (4, 2, 3). If the speed of the bird is 10 m/s, then the velocity vector of the bird is

(a) $5(\hat{i} - 2\hat{j} - 3\hat{k})$ (b) $5(4\hat{i} - 2\hat{j} - 3\hat{k})$

(c) $0.6\hat{i} - 0.8\hat{j}$ (d*) $6\hat{i} - 8\hat{j}$

3 unsoved Ex

7. A particle is moving in x-y-plane at 2 m/s along x-axis. 2 seconds later, its velocity is 4 m/s in a direction making 60° with positive x-axis. Its average acceleration for this period of motion is

(a) $5\ m/s^2$, along $y-$ axis

(b*) $3\ m/s^2$, along y - axis

(c) $5\ m/s^2$, along at 60° with positive $x-$ axis

(d) $3\ m/s^2$ at 60° with positive x - axis

5

8. A body covers first ½ part of his journey with a velocity of 2 m/s, next 1/3 part with a velocity of 3 m/s and rest of the journey with a velocity 6 m/s. The average velocity of the body will be

(a*) 3 m/s

(b) (11/3) m/s

(c) (8/3) m/s

(d) (4/3) m/s

6

9. A body starts from rest and is uniformly accelerated for 30s. The distance travelled in the first 10s is x_1, next 10s is x_2 and the last 10s in x_3. Then $x_1 : x_2 : x_3$ is the same as

(a) 1 : 2 : 4 (b) 1 : 2 : 5

(c*) 1 : 3 : 5 (d) 1 : 3 : 9

10

10. A stone is released from an elevator going up with an acceleration a and speed u. The acceleration and speed of the stone just after the release is

(a) a upward, zero

(b) $(g - a)$upward, u

(c) $(g - a)$downward, zero

(d*) g downward, u

12

11. Two particles held at different heights a and b above the ground are allowed to fall from rest. The ratio of their velocities on reaching the ground is

(a) a : b

(b*) $a : b$

(c) $a^2 : b^2$

(d) $a^3 : b^3$

14

12. A ball is dropped from a height. If it takes 2sec to cross the last 60 m before hitting the ground, find the height from which it was dropped

(a*) 80 m (b) 100 m

(c) 120 m (d) 180 m

20

13. A ball is projected upward from the foot of a tower. The ball crosses the top of the tower twice after an interval of 6s and the ball reaches the ground after 12s. The height of the tower is ($g = 10\ m/s^2$)

(a) 120 m (b*) 135 m

(c) 175 m (d) 80 m

22

14. Figure shows the position time graph of a particle moving on the x-axis

(a) The particle is continuously going in positive x direction

(b) Area under x – t curve shows the displacement of particle

(c) The velocity increases upto a time t_0 and then becomes constant

(d*) The particles moves at a constant velocity upto a time t_0 and then stops

24

15. In the displacement– time graph of a moving particle is shown, the instantancous velocity of the velocity is negative at the point

(a) C (b) D

(c*) E (d) F

25

16. The displacement time graphs of two particles A and B are straight lines making angles of respectively 30° and 60° with the time axis. If the velocity of A is V_A and that of B is V_B, then the value of V_A / V_B is

(a) 1/2 (b) $\frac{1}{3}$

(c) $\sqrt{3}$ (d*) 1/3

26

17. The variation of the velocity of a particle moving along the straight line is shown in the figure. The distance travelled by the particle in 4s is

(a) 25 m (b) 30 m

(c*) 55 m (d) 60 m

31

18. If the velocity v of a particle moving along a straight line decreases linearly with its displacement s from $20 ms^{-1}$ to a value approaching zero at s = 30 m, then acceleration of the particle at s = 15 m is

(a) $\frac{2}{3} ms^{-2}$ (b) $-\frac{2}{3} ms^{-2}$

(c) $\frac{20}{3} ms^{-2}$ (d*) $-\frac{20}{3} ms^{-2}$

35

19. A body is moving in a straight line as shown in the velocity time graph. The displacement and distance travelled by body in 8s are respectively

(a*) 12m, 20m (b) 20m, 12m

(c) 12m, 12m (d) 20m, 20m

37

20. The acceleration of a train between two stations 2 km apart is shown in the figure. The maximum speed of the train is

91. The volume of one mole of water at 277 K is 18 mL. One mL of water contains 20 drops. The number of molecules in one drop of water will be

(a) 1.07×10^{21}

(b*) 1.67×10^{21}

(c) 2.67×10^{21}

(d) 1.67×10^{20}

22

92. One atomic mass unit in kilogram is

(a) 1/NA (b) 12 / NA

(c*) 1/1000 NA (d) 1000 / NA

1

93. The number of g-atoms of nitrogen in its 7 g is equal to number of g-atoms in

(a) 6 g Mg (b*) 28 g Fe

(c) 30 g Ca (d) 20 g Hg

2

94. 2 isotopes of an element are present in 1 : 2 ratio of number, having mass number M and (M + 0.5) respectively. The mean mass number of element will be

(a) 3M + 1

(b*) 1.5 M + 0.5

(c) 0.5 M + 0.5

(d) None of these

3

95. A sample of nitrogen contains 90% N^{14} and 10% N^{15} isotopes. The average number of neutron per atom is

(a) 14.10 (b) 7.05

(c*) 7.10 (d) 14.20

4

96. From 2 mg calcium, 1.2×10^{19} atoms are removed. The number of g-atoms of calcium left is

(a) 5×10^{-5} (b) 2×10^{-5}

(c*) 3×10^{-5} (d) 5×10^{-6}

5

97. Three isotopes of an element have mass numbers M, (M+1) and (M+2). If the mean mass number is (M + 0.5), then which of the following ratio of number may be accepted for M, (M+1) and (M+2) in the order

(a) 1 : 1 : 1 (b*) 4 : 1 : 1

(c) 3 : 2 : 1 (d) 2 : 1 : 1

6

98. The number of electron in 3.1 mg NO_3^- is

(a) 32 (b) 1.6×10^{3}

(c*) 9.6×10^{20} (d) 9.6×10^{23}

7

99. The number of neutrons in 0.45 g water, assuming that all the hydrogen atoms are H1 atoms and all the oxygen atoms are O16 atoms, is

(a) 8 (b) 0.2

(c*) 1.2×10^{23} (d) 4.8×10^{24}

8

100. Number of oxygen atoms present is 1.61 g of $Na_4SO_4.10H_2O$ is

(a) 3.0 g C_2H_6 (b) 18.0 g $C_6H_{12}O_6$

(c*) 9.8 g H_2SO_4 (d) 8.0 g SO_3

10

PART-III BIOLOGY

PART-I BOTANY

Section A

101. In Whittaker's classification, which kingdom is main producer?
(a) Monera (b*) Protista
(c) Plantae (d) Mycota

102. Which statement is wrong–
(a*)Both Ascospores and Basidiospores are sexual spores formed endogenously
(b) Phenetic classification is based on observable characters of organisms
(c) Monograph contains information of any taxon
(d) Herbarium is a store house of collected plant specimens that are dried, pressed and preserved on sheets.

103. In the five-kingdom classification, Chlamydomonas and Chlorella have been included in :
(a*)Protista (b) Algae
(c) Plantae (d) Monera

104.According five kingdom classification of Whittaker, Saprophytic slime moulds are placed in
(a) Kingdom Fungi
(b) Kingdom Monera
(c*) Kingdom Protista
(d) Kingdom Animalia

105. During conjugation, bacteria attach by means of.
(a) Flagella (b*) Sex Pili
(c) Cilia (d) Pili

106. Ribosomes of Nostoc are
(a) 18 S (b) 16 S
(c*) 70 S (d) 80 S.

107. All bacterial cells get stained with
(a) Mercuric chloride
(b*)Crystal violet
(c) Crystal violet + iodine
(d) Safranin

108. Mucopeptide is abundant in cell wall of
(a) Cyanobacteria
(b*) Gram (+) bacteria
(c) Gram (-) bacteria
(d) Bacteriophage

109. Bacteria obtaining energy by oxidation of inorganic substances are
(a*) Chemo-lithotrophs
(b) Chemo-organotrophs
(c) Photo-lithotrophs
(d) Photo-organotrophs

110. Which one does not evolve oxygen
(a*)Photosynthetic bacteria
(b) Blue Green algae
(c) Green algae
(d) Autotrophic plants

111. Prokaryotic genetic material is
(a) Linear DNA + histones
(b) Circular DNA + histones
(c) Linear DNA without histones
(d*) Circular DNA without histones

112. Bacterial photosynthesis differs from photosynthesis of others in
(a) First product
(b) Number of phases
(c*) Type of reductant
(d) All the above

113. Match items of column I with those of column II and choose the correct combinations

Column-I	**Column-II**
(a) Pneumonia	(p) Vibrio
(b) Citrus Canker	(q) Mycobacterium leprae
(c) Cholera	(r) Yersinia pestis
(d) Leprosy	(s) Xanthomonas citri
	(t) Diplococcus pneumoniae

(a*)a-t, b-s, c-p, d-q
(b) a-t, b-p , c-s , d-q
(c) a-t, b-s, c-q, d-p
(d) a-t, b-q, c-p, d-s

114. The bacterium (*Clostridium botulinum*) that causes botulism is
(a) Obligate aerobe
(b) Facultative aerobe
(c) Facultative anaerobe
(d*)Obligate anaerobe

115. The most primitive monerans are
(a) Rickettsiae
(b) Actinomycetes
(c) Progenote
(d*)Archaebacteria

116. Ti gene occurs in
(a) *Bacillus thuringiensis*
(b) *Escherichia coli*
(c*)*Agrobacterium tumefaciens*
(d) *Rhizobium leguminosarum*

117. Which of the following statement is/are correct with respect to Bacteria
(a) Bacteria are the sole members of the Kingdom Monera
(b) Bacteria occur almost everywhere
(c) Bacteria are the most abundant micro organism
(d) 80 S type of Ribosome are present in bacteria
(a) a & d (b) b & d
(c*)b, c (d) b, c, d

118. Contractile vacuole of*Euglena* is
(a) Posterior
(b) Middle
(c) Anterior
(d*)Anterior and fixed

119. Match the column I with Column II

Coumn I	Column II
(i) Presence of branched chain lipids in cell membrane	(a) Thermo acidophiles
(ii) Homopolar bond in protein	(b) Methanogens
(iii) Found in marshy habitats, swamps, ruminants, sewage treatment plants	(c) Halophiles

(a) (i) a, (ii) b, (iii) c
(b) (i) a, (ii) c, (iii) a
(c) (i) b, (ii) c, (iii) a
(d*)(i) c, (ii) a, (iii) b

120. (i) Monera Archaebacteria, Eubacteria, BGA, Euglenoids
(ii) Protista Diatoms, Dinoflagellates, Euglenoids
(iii) Fungi *Albugo*, *Mucor*, Penicillium, Slime-moulds
Which one is/are correct match
(a) Only (i) (b*) Only (ii)
(c) Only (iii) (d) All are correct

121. The diatomaceous earth is used for insulating boilers and steam pipes because
(a) The diatomaceous earth is very cheap
(b) It is a good conductor of heat
(c*)It is a bad conductor of heat
(d) It is composed of calcium carbonate

122. Protists which are diploid reproduce sexually by the process of
(a) Zygotic meiosis
(b) Cyst formation
(c) Binary fission
(d*)Gametic meiosis

123. Cell wall of Diatoms is made up
(a*)Silica
(b) Carbonate
(c) Calcium carbonate
(d) All of these

124. Formation of diatomaceous earth has occurred due to remains of the following part of diatoms
(a*)Cell wall
(b) Cytoplasm
(c) Chloroplast
(d) None of the above

125. PSP (Paralytic shellfish poisoning) is connected with
(a*)*Gonyaulax*
(b) *Ceratium*
(c) *Noctiluca*
(d) *Glenodinium*

126. Bioluminescence was first recorded in
(a) *Gonyaulax*
(b) *Peridinium*
(c*)*Noctiluca*
(d) *Pyrodinium*

127. Fire algae are members of
(a) Phaeophyceae
(b*)Dinophyceae
(c) Rhodophyceae
(d) Bacillariophyceae

128. Dinophycean theca is made of
(a) Silica (b) Lime
(c*)Cellulose (d) Protein

129. Paramylum granules are made of
(a*)□ 1-3 glucan
(b) □ 1-3 glucan
(c) □ 1-4 glucan
(d) □ 1-4 glucan

130. Eye spot granules of*Euglena* are formed of
(a) Fucoxanthin
(b*)Astaxanthin
(c) □ -carotene
(d) □-carotene

131. Cod liver oil is derived from
(a*)Diatoms
(b) Dinoflagellates
(c) Euglenoids
(d) All the above

132. Eye spot of*Euglena* is a
(a*)Nonplastid structure found near the reservoir
(b) Nonplastid structure found on the side near the middle
(c) Plastid structure present near the reservoir
(d) Plastid structure found near the middle.

133. The storage product of chrysophytes is
(a*)Oil and laminarin
(b) Starch
(c) Paramylum
(d) Glycogen

134. Photosynthetic protists belong to
(a) Bacillariophyceae
(b) Bacilariophyceaeand euglenophyceae
(c*)Bacillariophyceae,euglenophyceae and dinophyceae
(d) Zooflagellates.

135. Protistans are connecting link between
(a) Plants and animals
(b) Fungi and plants
(c*)Monerans and kingdoms of multicellular organisms
(d) Fungi and animals.

Section B

136. Percentage of global photosynthesis performed by photosynthetic protists is
(a) 30% (b) 50%
(c) 65% (d*) 80%

137. Which statement is/are**wrong** with respect to kingdom protista.
(i) All single celled eukaryotes are placed under protista
(ii) The boundaries of this kingdom are not well defined.
(iii)Chrysophytes, dinoflagellated, Euglenoids, slimemoulds and protozoans are included under protista.
(iv) All protista are eukaryotic, achlorophyllous, heterotrophic, nonvascular organism.
(a) only (ii) (b) only (iii)
(c*) only (iv) (d) None of these

138. Slime moulds are _____(i)_____protists. The body moves along decaying twigs and leaves engulfing organic material under suitable conditions, they form an aggregation called_____(ii)_____which may grow and spread over several feet.
In above question (i) & (ii) are respectively.
(a) (i) Autotrophic, (ii)plasmodium
(b) (i) Chemosynthetic autotrophic (ii) plasmodium
(c) (i) Saprophytic, (ii) Phycobiont
(d*)(i) saprophytic, (ii)plasmodium

139. Read statements A to D
(A) Golden Algae are classified under chrysophytes.
(B) *Gonyaulax* & Diatoms are classified under Dinoflagellates
(C) In Diatoms the cell wall form thin overlapping shell, which fit together as in soap box.

(D) Most of the Dinoflagellates have two flagella.
How many statement are wrong.
(a*)one (b) two
(c) three (d) four

140. Read the following four statements (A-D) and answer as asked next to them.
(A) All single celled eukaryotes are placed under protista
(B) Most of the organism of chrysophytes are photosynthetic.
(C) Dinoflagellates appear yellow, green brown, blue or red depend on the main pigments present in their cells.
(D) Chloroplast absent in Euglena
How many of the above statement are correct ?
(a) Two (b*) Three
(c) Four (d) one

141. Euglena is considered as connecting link between plant Kingdom & animal kingdom because it shows features of both plants & animals which of the following is/are plant character of Euglena .
(A) Presence of chlorophyll pigment
(B) Holozoic nutrition
(C) Holophytic nutrition
(D) Presence of sexual reproduction.
(a) A & B (b) B & C
(c) C & D (d*) A & C

142. Select the wrong pair
(a) Red tide – Gonyaulax
(b*)Cellulosic cell wall – Archaebacteria
(c) Saprophytic protist – Slime moulds
(d) Mycoplasma – Cell wall less & can survive without oxygen.

143. What is lacking in fungi
(a) Carotenoids
(b*)Chlorophyll
(c) Both (a) & (b)
(d) None of these

144. Mesokaryon is
(a) A nucleus like structure
(b) Nucleoid with condensed chromosomes
(c*) A nucleus with condensed chromosomes
(d) A nucleoid with distinct chromosomes.

145. In dinophyceae the two flagella are
(a) Anterior
(b*)One transverse and other vertical
(c) Lateral
(d) Posterior

146. The fungi growing on faecal matter are called
(a*)Coprophilous
(b) Saprophyte
(c) Parasite
(d) Symbiont

147. Thick walled structure having reserve food which fulfils the function of vegetation reproduction is called as
(a) Bud
(b) Oidia
(c*) Chlamydospore
(d) Torula

148. Where the basidiospores are formed
(a) In basidium (Endogenous)
(b*)On basidium (exogenous)
(c) In basidiocarp
(d) On basidiocarp

149. In which of the following fungal groups the cell wall is made up of cellulose
(a) Zygomycetes
(b) Trichomycetes
(c*)Phycomycetes
(d) Ascomycet

150. Which of these is used in making wine
(a*)Yeast (b) Bacteria
(c) Both of above (d) False yeast

PART ¯II (ZOOLOGY)
Section A

PT ¯4 class XI

151. Mark the correct match of the animal and its common name
(a) Trygon– dog fish
(b) Ascidia– lancelet
(c) Pterophyllum– flying fish
(d*) Myxine– hagfish

152

152. Which of the following is a limbless amphibians?
(a) Salamander
(b*) Ichthyophis
(c) Pristis
(d) Cucumaria

153

153. Which of the following is the exclusive feature of the birds?
(a) 4-chambered heart
(b) Homeothermic
(c) Fore limbs modified into wings
(d*) Presence of feathers

154

154. In which one of the following features crocodile and penguin are similar to whale and dogfish
(a) Possess a solid single central nervous system
(b) Lay eggs and guard them till they hatch
(c) Possess external ear
(d*) Have gill slits at some stages

156

155. Lamprey is not a fish since it does not have
(a) Closed blood vascular system
(b) Fins
(c*) Body scales and paired fins
(d) Cranium and vertebral column

159

156. In chordates the notochord is
(a) Mesodermal and dorsal to nerve chord
(b) Endodermal and dorsal to nerve chord
(c*) Mesodermal and ventral to nerve chord
(d) Endodermal and ventral to nerve chord

161

157. The petromyzones are
(a) Only marine and non-migratory
(b) Only fresh water form and non-migratory
(c*) Marine and migrate to fresh water for spawning
(d) Fresh water form and migrate to sea for spawning

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158. Which of the following character is not true for cyclostomata?
(a) Ectoparasites on some fishes
(b) Bearings 6-15 pairs of gill stins
(c) Sucking and circular mouth without jaws
(d*) Cranium and vertebral column are bony

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159. Which of the following have placoid scales and due to the absence of air bladder, they have to swim constantly to avoid sinking
(a) Labeo
(b) Exocoetus
(c) Hippocampus
(d*) Pristis

166

160. Which of the following have electric organs?
(a*) Torpedo
(b) Trygon
(c) Scoliodon
(d) Carcharodon

167

161. In which of the following external fertilization occurs and are mostly oviparous?
(a*) Osteichthyes
(b) Chondrichthyes
(c) Reptiles
(d) All of these

168

162. In birds scales are generally present on
(a) Forelimbs
(b*) Hindlimbs
(c) Beak
(d) Whole skin

171

163. Which of the following characters of chordate are not correct from given characters?
i. Protochordates are exclusively marine
ii. In urochordata notochord is present only in larval tail
iii. In cephalochordate notochord extends from head to tail region and is persistent throughout their life
iv. In vertebrates notochord is replaced by a cartilaginous or bony vertebral column in the embryo
v. All living members of the class cyclostomata are endoparasites in some fishes
(a) i, ii and iii
(b) i, iv and v
(c*) iv and v
(d) Only v

172

164. Which characters for cyclostomata are incorrect from given characters?
i. They have an enlongated body, bearing 6-15 pairs of gill slits
ii. Their body is devoid of scales and paired fins
iii. Cranium and vertebral column are cartilaginous
iv. Cyclostomes are fresh water animals but migrate for spawning to marine
v. After migrate for spawning to marine water within a few days they die
vi. Their larvae after metamorphosis return to the river
(a) i, ii, iii, iv and v
(b) iii, iv, v and vi

(c) iv and v
(d*) iv, v and vi

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165. Which of the following is not true statement for bony fishes?
i. Pristis, carcharodon and trygon are members of it
ii. It includes both marine and fresh water fishes
iii. They have four pairs of gills which covered by operculum
iv. Skin is covered with placoid scales
v. Sexes are separate and fertilization is usually external
(a) i, ii, iii and iv
(b*) i and iv
(c) ii, iv and v
(d) i, ii and iii

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166. Which of the following statements are incorrect for Reptilia?
i. Salamandra calotes and ichthyophis are members of reptilian
ii. Their body is covered by dry and cornified skin, epidermal scales and scutes
iii. They do not have external ear openings and tympanum represents ear
iv. Snakes and lizards shed their scales as skin cast
v. Heart is usually three-chambered but four-chambered in crocodiles
vi. Naja, bangarus, vipera are non-poisonous snakes
(a) i, ii, iii and iv
(b) ii, iii and v
(c) i, ii and iv
(d*) iii, v and iv

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167. How many statements are correct from given statements?
i. The hindlimbs of birds have generally scales
ii. Skin of bird is dry without glands except the oil gland at the base of the tail
iii. Long bones of the mammals are hollow with air cavities
iv. Reptiles, aves and mammals are homoiothermous animals
v. Ornithorhynchus, Macropus and Pteropus are member of mammals
(a) One (b*) Three
(c) Four (d) Five

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168. Which one of the following is not a poisonous snake?
(a) Cobra (b) Viper
(c*) Python (d) Krait

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Given below are assertion and reasons mark the correct choice from given options
(a) Assertion and reason both are true and the reason is correct explanation of assertion
(b) Assertion and Reason both are true but the reason is not a correct explanation of the assertion
(c) Assertion is true but the Reason is wrong
(d) Assertion and Reason are wrong

169. Assertion : Lamprey and Hagfish are vertebrates
Reason : Posses notochord during the embryonic period
(a) (b*)
(c) (d)

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170. Assertion : All vertebrates are chordates but all chordates are not vertebrates
Reason : The notochord is replaced by a cartilaginous or bony vertebral column in the vertebrates only
(a*) (b)
(c) (d)

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171. Assertion : Psittacula and struthio are member of aves
Reason : The forelimbs are modified into wings and both of them can fly
(a) (b)
(c*) (d)

183

172. Assertion : Ornithorhynchus and macropus are member of mammalia
Reason : They have milk producing glands
(a*) (b)
(c) (d)

184

173. Which one of the following categories of animals is correctly described with no single exception in it?
(a*) All bony fishes have four pairs of gills and an operculum on each side
(b) All sponges are marine and have collared cells
(c) All mammals are viviparous and possess diaphragm for breathing
(d) All reptiles possess scales have a three chambered heart and are cold blood

190

174. Match the name of the animal (column I) with characteristics (column II) and the phylum/class (column III) to which it belongs

	Column I	**Column II**	**Column III**
(a)	Icthyophi	Terrestrial	Reptile
(b)	Limulus	Body covered by Chitinous exoskeleton	Pisces
(c)	Adamsia	Radially Symmetrical	Porifera
(d*)	Petromyzon	Ectoparasite	cyclostomata

191

175. Which of the following represents the correct combination without any exception?

Characters	**Class**
(a) Mouth ventral Gills without Operculum, skin With placoid scales Persistent notochord	Chondrichthyes
(b) Sucking and Circular mouth; Jaws absent Integument without Scales, paired appendages	Cyclostomata
(c) Body covered With feathers; skin Moist and glandular Forelimbs from Wings; lungs with Air sacs	Aves
(d) Mammary Gland; hair on Pinna;two pairs Of limbs	Mammalia

192

176. Which one of the following characteristics is not shared by birds and mammals?

(a) Ossified endoskeleton
(b) Breathing using lungs
(c*) Viviparity
(d) Warm-blooded nature

193

177. Choose the correct statement

(a) All reptiles have a three chambered heart
(b) All pisces have gills covered by an operculum
(c) All mammals are viviparous
(d*) All cyclostomes do not possess jaws and paired fins

194

178. Which one of these animals is not a homeotherm?

(a) Macropus (b*) Chelone
(c) Camelus (d) Psittacula

195

179. Which of the following characteristics features always holds true for the corresponding group of animals

(a*) Cartilaginous endoskeleton	Chondricthyes
(b)Viviparous	Mammalia
(c) Possess a Mouth with an Upper and lower jaw	Chordata
(d) 3-chambered heart With one incompletely Divided ventrical	Reptilia

196

180. Which of the following statements are true for the phylum-chordata?

i. In urochordata notochord extends from the head to tail and it is present throughout their life
ii. In vertebrata notochord is present during the embryonic period only
iii. Central nervous system is dorsal and hollow
iv. Chordata is divided into 3 subphylia Hemichordata, tunicate and cephalochordate

(a) iii and i (b) i and ii
(c*) ii and iii (d) iv and iii

197

181. In which of the following animals digestive tract has additional chambers like crop and gizzard

(a) Catla, Columba, crocodiles
(b*) Pav, psitacula, corvus
(c) Corvus, Columba, chameleon
(d) Bufo, Balaenoptera, bangarus

198

182. Which one of the following features are not true for chordates?

i. Heart is dorsal
ii.Pharynx is perforated by gill shits
iii.Central nervous system is ventral, solid and single
iv.Post anal tail is present
v.Notochord is present

choose the most appropriate answer from the options given below

(a*) i and iii only (b) v only
(c) i, iv and v only (d) ii and iii only

199

183. Given below are two statements

Statement I : Amphibians and reptiles have a 3-chambered heart with two atria and single ventricle and oviparous in natute

Statement II : Crocodiles possess a 4-chambered heart with two ventricles and two atria and viviparous in natute

In the light of the above statements choose the most appropriate answer from the options given below

(a) Both statement I and II are incorrect
(b*) Statement I is correct but statement II is incorrect
(c) Statement I is incorrect but statement II is correct
(d) Both Statement I and II are correct

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184. Which one of the following is a matching pair of a body feature and the animal possessing it?

(a) Ventral heart - scorpion
(b) Post-anal– tail-cucomeria
(c*) Ventral central nervous system - Leech
(d) Pharyngeal gill slits in embryo - chameleon

185. What is true about nereis, scorpion, cockroach and silver fish
(a) They all belong to the same phylum
(b) They all have jointed paired appendages
(c*) They all possess dorsal heart
(d) None of them is aquatic
163

Section-B

186. Which of the following phyla is correctly matched with its two general characteristics?
(a) Arthropoda : Body is divided into head, thorax
and abdomen and respiration by the lungs
(b) Chordata : Notochord at some stage and separate anal and urinary openings to the outside
(c) Echinodermata : Pentamerous radial symmetry and mostly internal fertilisation
(d*) Mollusca : Mostly oviparous and development through a trochophore or veliger larva
164

187. Ascaris is characterised by
(a) Absence of true coelom but presence of metamerism
(b*) Presence of neither true coelom nor metamerism
(c) Presence of true coelom but absence of metamerism
(d) Presence of true coelom and metamerism
165

188 How many statements are true for level of organisation of animals
i. Sponges exhibit cellular level of organisation
ii. Coelenterates exhibit tissue level of organisation
iii. Organ level is exhibited by members of Platyhelminthes
iv. Annelids, Arthropods, Molluscus, Echinoderms and chordates exhibit organ system level of organisation
v. Hemichordates exhibit organ level organisation
(a) One (b) Three
(c*) Four (d) Five
167

189. How many characters for mollusca are correct from given characters
i. Body have distinct head, muscular foot ans visceral hump
ii. A soft and spongy layer of skin forms a mantle over the visceral hump
iii. The space between the head and the mantle is called the mantle cavity
iv. Mantle cavity have feather like gills which related with respiratory and excretory function
v. The mouth contains a file-like rasping organ for feeding called radula
vi. They are usually monoecious and oviparous with indirect development
(a*) i, ii, iv and v (b) ii, iii and vi
(c) I, ii, iii and vi (d) I, ii, iii, iv and vi
168

190. How many characters for Echinodermata are correct from given characters
i. These animals have an exoskeleton of calcareous ossicles
ii. All are marine with organ-system level of organisation
iii. Digestive system is complete with mouth on the dorsal side and anus on the ventral side
iv. Water canal system present which helps in locomotion, capture and transport of food and respiration
v. They are monoecious, fertilisation usually external
vi. Development is indirect with free-swimming larva
(a) ii, iv, v and vi (b) i, ii, iii, v and vi
(c) I, ii, iv, v and vi (d*) ii and vi
169

191. Choose the correct one for hemichordate from given statements
i. The body is cylindrical and is composed of an anterior proboscis a collar and a lung trunk
ii. Circulatory system is of open type
iii. Respiration takes place through gills
iv. Excretory organ is proboscis gland
v. They are dioecious fertilisation is external and development is indirect
(a) i, iii and iv only (b) i, ii, iv and v only
(c*) i, ii, iii and vi (d) ii and vi
170

192. Which of the following option is correct for given diagram?



(a) Balanoglossus-close circulation
(b) Balanoglossus-respiration by proboscis gland
(c*) Balanoglossus-open circulation
(d) Nereis-open circulation
172

193. Match the column I with column II and choose the correct option

Column I	**Column II**
A. Porifera	**i.** Canal system
B. Aschelminthes	**ii.** Water vascular system
C. Annelida	**iii.** Muscular Pharynx
D. Arthropoda	**iv.** Comb plates
E. Echinodermata	**v.** Metameres
	vi. Joined appendages

(a) A-ii, B-iii, C-v, D-iv, E-i
(b) A-ii, B-v, C-iii, D-iv, E-ii
(c*) A-i, B-iii, C-v, D-vi, E-ii
(d) A-i, B-v, C-iii, D-iv, E-ii
176

Given below are assertion and reasons mark the correct choice from given options

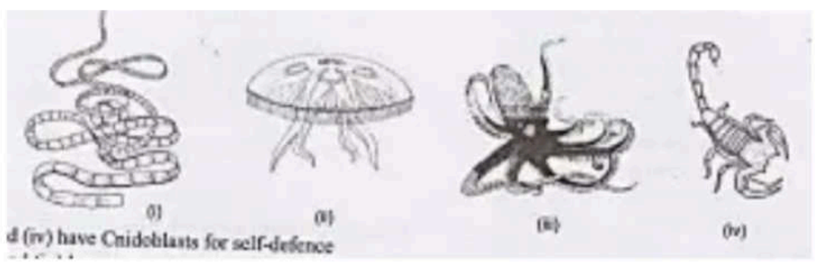
(a) Assertion and reason both are true and the reason is correct explanation of assertion
(b) Assertion and Reason both are true but the reason is not a correct explanation of the assertion
(c) Assertion is true but the Reason is wrong
(d) Assertion and Reason are wrong

194. Assertion : Arthropods are the largest phylum of animals which includes arthropods
Reason : Over two thirds of all named species on earth are arthropods
(a*) (b)
(c) (d)
179

195. Assertion : Hemichordata was earlier considered as a sub-phylum under the phylum chordata
Reason : Balanoglossus and saccoglossus are urochordates
(a) (b)
(c*) (d)
180

196. Budding is a common method of asexual reproduction. It is found in
(a*) Hydra and sponge
(b) Star fish and hydra
(c) Tapeworm and hydra
(d) Sponge and star fish
181

197. The fig, shows four animals I, ii, iii and iv. Select the correct answer with respect to a common characteristics of two of these animals



(a) i and iv have cnidoblasts
(b*) iii and iv have a true coelum
(c) i and iv respire mainly through body wall
(d) ii and iii show radial symmetry
187

198. Select the taxon mentioned that represents both marine and fresh water species
(a) Cephalochordata (b*) Cnidaria
(c) Echinoderms (d) Ctenophora
188

199. Body having mesh work of cells, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum
(a) Protozoa
(b) Coelenterata
(c*) Porifera
(d) Mollusca
190

200. Consider the following features

A. Organ system level of organisations
B. Bilateral symmetry
C. True coelomates with segmentation of body

Select the correct option of animal groups which possess all the above characteristics
(a) Annelida, Arthropoda and Mollusca
(b) Arthropoda, Mollusca and Chordata
(c) Annelida, Mollusca and Chordata
(d*) Annelida, Arthropoda and Chordata
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